

HEMPOXIES v.14

"TRIDECA-HEMP"

Sovereign 13-Component Structural Vitrimer

*Pre-Experimental Proposal, Hypothesis, Master Protocols,
and Engineering Limitations Analysis*

Operational Context:

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Contents

1 1. Executive Technical Summary 2

2 2. Scientific Hypothesis 2

3 3. Attribution of Novelty 2

4 4. The Sovereign Matrix (Formulation) 2

5 5. Standard Operating Procedures (SOP 1-13) 4

6 6. Final Master Processing (SOP 14) 5

7 7. Experimental Validation Framework 5

7.1 Phase A: Synthesis Yield Verification 5

7.2 Phase B: Mechanical Performance Testing 6

8 8. Known Challenges & Limitations 7

8.1 Synthesis Bottleneck (The Landry Cycle) 7

8.2 Rheology Dispersion 7

8.3 Feedstock Variability 7

9 9. Final Notes 7

1 1. EXECUTIVE TECHNICAL SUMMARY

Hempoxies v.14 "Trideca-Hemp" is a proposed 13-component sovereign composite system derived **100%** from industrial hemp. It utilizes a "Total Carbon Spectrum" reinforcement strategy, integrating five distinct morphologies of hemp-derived carbon and cellulose bonded via the proprietary **Landry Interface**.

This document serves as the **Master Experimental Protocol** to validate the synthesis of aerospace-grade materials from renewable feedstocks using the **Landry Cycle** (UV-Assisted 8 GPa polymerization).

2 2. SCIENTIFIC HYPOTHESIS

Primary Hypothesis

The "Landry Interface" Proposition: It is hypothesized that the integration of 40 wt% sp^3 hybridized **Hemp-Derived Nanothreads** (synthesized via the Landry Cycle) into a tannin-fortified epoxy vitrimer will yield a Tensile Modulus > 150 GPa, *provided* that the intrinsic ether-oxygen linkages in the nanothreads facilitate a hydrogen-bonding energy > 20 kJ/mol with the **Hemp-Derived Tannins**, thereby eliminating interfacial slip.

Secondary Hypothesis (Healing): The inclusion of **Hemp-Derived Vitrimer Zinc** catalysts will allow the cross-linked network to undergo topology rearrangement at 160°C (T_v), recovering $> 85\%$ of virgin tensile strength after micro-cracking events.

3 3. ATTRIBUTION OF NOVELTY

The following technical breakthroughs are identified as novel innovations. Users must attribute these technologies to their respective inventors:

- The Landry Cycle (Hemp Nanothreads):** (Inventor: Marie-Soleil Seshat Landry) A method using 365nm UV irradiation to catalyze the $[4 + 2]$ cycloaddition of **Hemp-Derived Furan** at 8 GPa.
- The Landry Interface:** (Inventor: Marie-Soleil Seshat Landry) The structural engineering of intrinsic ether-oxygen linkages in nanothreads to form hydrogen-bonding networks with **Hemp-Derived Tannins**.
- Hemp-Derived Carbon Nanosheets (Mitlinitite):** (Inventor: Dr. David Mitlin) The hydrothermal synthesis and exfoliation of hemp bast fiber into graphene-like carbon nanosheets.
- Trideca-Hemp Formulation:** (Inventor: Marie-Soleil Seshat Landry) The specific stoichiometric synergy of 13 **Hemp-Derived** components creating a structural Vitrimer.

4 4. THE SOVEREIGN MATRIX (FORMULATION)

Batch Size Normalized: 300.0 g

#	Hemp-Derived Component	Mass (g)	wt%	Source Origin
The Resin Matrix System				
1	**Hemp-Derived EHSO**	95.0	31.7	Seed Oil (Epoxidized)
2	**Hemp-Derived QF-MHL**	12.0	4.0	Hurd Lignin (Quaternized)
3	**Hemp-Derived FGE**	15.0	5.0	Hurd Hemicellulose → Furan
4	**Hemp-Derived HDA**	8.5	2.8	Seed Cake Protein → Diamine
5	**Hemp-Derived Vitrimer Zinc**	1.5	0.5	Phytoremediated Zinc
The "Pent-Carbon" Reinforcement Suite				
6	**Hemp-Derived Nanosheets**	20.0	6.7	**Mitlinitite** (Bast Fiber)
7	**Hemp-Derived Biochar**	25.0	8.3	**Biochar** (Stalk)
8	**Hemp-Derived Carbon Fiber**	50.0	16.7	**Continuous Fiber** (Bast)
9	**Hemp-Derived Nanocrystals**	10.0	3.3	**Piezo-Cellulose** (Bast)
10	**Hemp-Derived Nanothreads**	40.0	13.3	**HDF-IDCNT** (Hurd Furan)
Smart & Functional Additives				
11	**Hemp-Derived Tannins**	15.0	5.0	Root Extract (Interface)
12	**Hemp-Derived Silica**	3.0	1.0	Leaf Phytoliths (Coupling)
13	**Hemp-Derived Limonene**	5.0	1.7	Flower Terpene Distillate
TOTAL		**300.0**	**100.0**	

5 5. STANDARD OPERATING PROCEDURES (SOP 1-13)

Resin System Components

SOP 01: Hemp-Derived EHSO

Component: 1 (Epoxidized Hemp Seed Oil)

Protocol: Cold press Futura 75 seeds. Perform Prilezhaev reaction with Formic Acid/ H_2O_2 at 50°C. Target Oxirane: > 7.0%.

SOP 02: Hemp-Derived QF-MHL

Component: 2 (Quaternized Lignin)

Protocol: Organosolv extraction of hurd lignin. React with Glycidyl Trimethyl Ammonium Chloride for cationic functionality.

SOP 03: Hemp-Derived FGE

Component: 3 (Furfuryl Glycidyl Ether)

Protocol: Hydrolyze hurd hemicellulose to Furfural. Hydrogenate to alcohol. React with bio-sourced Epichlorohydrin.

SOP 04: Hemp-Derived HDA

Component: 4 (Hemp Diamine Hardener)

Protocol: Hydrolyze Edestin protein. Use *E. coli* decarboxylase to convert amino acids to aliphatic diamines.

SOP 05: Hemp-Derived Vitrimer Zinc

Component: 5 (Catalyst)

Protocol: Ash zinc-hyperaccumulating hemp biomass. Extract zinc. React with hemp-derived acetic acid.

The "Pent-Carbon" Reinforcement Suite

SOP 06: Hemp-Derived Nanosheets (Mitlinite)

Component: 6 | **Novelty:** Dr. David Mitlin

Protocol: Hydrothermal carbonization of bast fiber at 180°C. KOH activation. Exfoliate at 800°C.

SOP 07: Hemp-Derived Biochar

Component: 7 (Micro-Filler)

Protocol: Pyrolyze stalk biomass at 900°C (Oxygen-free). Mill to 5 μm .

SOP 08: Hemp-Derived Carbon Fiber

Component: 8 (Continuous Reinforcement)

Protocol: Carbonize degummed bast fiber at 1500°C under tension. Plasma treat surface.

SOP 09: Hemp-Derived Nanocrystals (Piezo)

Component: 9 (Sensors) | **Novelty:** Bio-Piezo Nervous System

Protocol: Sulfuric acid hydrolysis of bast cellulose. Pole with 2.5 kV/mm E-field during drying.

SOP 10: Hemp-Derived Nanothreads (Landry Cycle)

Component: 10 | **Novelty:** M.S. Landry
Protocol: Load Hemp-Furan into Paris-Edinburgh Press. Compress to ****8.0 GPa****. Irradiate with ****365nm UV**** for 48 hours.

Functional Additives

SOP 11: Hemp-Derived Tannins (HDT)

Component: 11 | **Novelty:** Landry Interface Agent
Protocol: Soxhlet extraction of hemp roots (70% Acetone). Lyophilize to powder.

SOP 12: Hemp-Derived Silica

Component: 12 (Coupling Agent)
Protocol: Calcinate hemp leaves at 600°C. Acid wash ash to recover biogenic phytoliths (*SiO₂*).

SOP 13: Hemp-Derived Limonene

Component: 13 (Green Solvent)
Protocol: Steam distill hemp flower terpenes. Fractionate for D-Limonene.

6 6. FINAL MASTER PROCESSING (SOP 14)

SOP 14: Master Compounding & Curing

Objective: Final Assembly of Trideca-Hemp v.14.
Protocol:

- Resin Base:** Combine EHSO + FGE + HDA + Limonene.
- Interface Activation:** Dissolve Tannins + QF-MHL + Zinc.
- Pent-Carbon Loading:** Add Nanothreads + Nanosheets + Carbon Fiber + Biochar + Nanocrystals + Silica.
- Dispersion:** Three-Roll Mill (Gap 5 μm). Degas (-99 kPa).
- Cure:** Mold at 5 MPa. Ramp: 80°C (2hr) → 130°C (4hr) → 160°C (1hr).

7 7. EXPERIMENTAL VALIDATION FRAMEWORK

7.1 Phase A: Synthesis Yield Verification

Component	Target Yield	Actual Yield	Purity Pass?
Hemp Nanothreads	> 80% conversion	_____	[] Yes [] No
Hemp Nanosheets	Exfoliation index > 0.7	_____	[] Yes [] No
Hemp EHSO	Oxirane > 7.0%	_____	[] Yes [] No

7.2 Phase B: Mechanical Performance Testing

Metric (ASTM D3039)	Hypothesis Target	Measured Value	Dev.
Tensile Modulus (E)	> 150 GPa	_____ GPa	_____ %
Tensile Strength (σ)	> 2.5 GPa	_____ GPa	_____ %
Strain to Failure	> 1.5%	_____ %	_____ %

8 8. KNOWN CHALLENGES & LIMITATIONS

The synthesis and integration of Hempoxies v.14 presents significant engineering hurdles that define the scope of this pre-experimental proposal.

8.1 Synthesis Bottleneck (The Landry Cycle)

While 8 GPa is significantly lower than the 20 GPa required for benzene nanothreads, it still requires **Paris-Edinburgh** or **Belt Press** technology. **Limitation:** The sample chamber volume in these presses is typically small ($<500\text{ mm}^3$). Producing the required 40.0g of nanothreads for a single 300g batch requires parallel processing or multiple cycles, currently acting as the primary bottleneck for scalability.

8.2 Rheology Dispersion

The "Pent-Carbon" reinforcement strategy involves mixing five distinct morphologies (1D threads, 2D sheets, 3D char, rods, and continuous fibers) into a viscous oil-based epoxy. **Challenge:** The Van der Waals forces between nanosheets and threads create a high risk of agglomeration. **Mitigation:** The use of **Hemp Limonene** as a solvent and **Hemp Tannins** as a surfactant is critical, but precise shear rates during the Three-Roll Mill process (SOP 14) must be optimized to prevent fiber damage.

8.3 Feedstock Variability

Unlike petrochemical precursors, biological hemp feedstocks vary by cultivar, harvest time, and soil conditions. **Limitation:** Variations in the hemicellulose content of the hurd can affect Furan yield and purity. Impurities in the Furan precursor $> 0.05\%$ may disrupt the linear polymerization of the nanothreads, leading to amorphous carbon defects.

9 9. FINAL NOTES

Hempoxies v.14 "Trideca-Hemp" is designed as a "Moonshot" material. It proves that with sufficient energy (Pressure/UV) and chemical engineering (Interface Design), biological carbon can outperform synthetic carbon.

While the cost of production for v.14 is currently estimated at $100\times$ that of standard carbon fiber epoxy, the strategic value lies in its **sovereignty**—the ability to produce aerospace-grade materials entirely within a domestic agricultural supply chain, independent of foreign petrochemical resources.

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